Heart Disease Data Integration and KNN Classification

Data Mining – Homework 1

Mahmoud Sadrian
Student ID: 40435336
GitHub Repository

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Abstract

This report presents a data mining study on the Heart Disease dataset from the UCI Machine Learning Repository. The main goal is to predict the presence of heart disease based on patients' clinical attributes using the K-Nearest Neighbors (KNN) classification algorithm. The data were preprocessed through handling missing values, normalizing numerical features, and splitting the dataset into training and testing sets. Exploratory data analysis (EDA) was conducted using histograms, boxplots, Q-Q plots, correlation matrices, pair plots, and count plots to better understand the feature distributions and relationships. The KNN classifier was implemented from scratch with a mixed distance function combining numerical and categorical attributes. The model performance was evaluated using accuracy, precision, recall, F1-score, and confusion matrix. The results show that KNN can achieve reasonable performance in predicting heart disease, demonstrating how simple distance-based methods can be applied to medical decision support.

1 Introduction

Cardiovascular diseases remain one of the leading causes of mortality worldwide. Early detection and risk assessment play a crucial role in preventing severe outcomes and improving patients' quality of life. In recent years, data-driven methods and machine learning techniques have become increasingly important tools for analyzing medical data and supporting clinical decision making.

In this project, we apply data mining techniques to the well-known Heart Disease dataset from the UCI Machine Learning Repository. The main objective is to investigate how patients' characteristics, such as age, sex, chest pain type, cholesterol level, resting blood pressure, and maximum heart rate, can be used to predict the presence of heart disease.

We focus on the K-Nearest Neighbors (KNN) classification algorithm, which is a simple yet powerful distance-based method. The project includes several main steps: data preprocessing, exploratory data analysis (EDA), implementation of the KNN classifier for mixed-type attributes, and performance evaluation using standard classification metrics. The results of this study illustrate the potential of KNN for medical classification tasks and highlight the importance of proper preprocessing and feature analysis in data mining workflows.

2 Dataset Description

The dataset used in this project is the Heart Disease dataset from the UCI Machine Learning Repository [1]. It consists of patient records collected from four different medical centers: Cleveland, Hungarian, Switzerland, and VA Long Beach. Each subset includes similar clinical attributes that describe the health status of patients and the presence or absence of heart disease.

In this work, we use the integrated version of the dataset, which contains approximately **1000** instances and **13** main attributes, along with a binary target variable indicating whether the patient is diagnosed with heart disease (`target = 1`) or not (`target = 0`). The attributes can be grouped into three categories:

- **Numeric (continuous) features**, such as:
  - `age`: age in years
  - `trestbps`: resting blood pressure (in mm Hg)
  - `chol`: serum cholesterol (in mg/dl)
  - `thalach`: maximum heart rate achieved
  - `oldpeak`: ST depression induced by exercise relative to rest
- **Binary or categorical features encoded as integers**, for example:
  - `sex`: sex (0 = female, 1 = male)
  - `cp`: chest pain type (0–3)
  - `fbs`: fasting blood sugar > 120 mg/dl (0 = false, 1 = true)
  - `restecg`: resting electrocardiographic results
  - `exang`: exercise induced angina (0 = no, 1 = yes)

- `slope`: the slope of the peak exercise ST segment
- `ca`: number of major vessels (0–3) colored by fluoroscopy
- `thal`: thalassemia (0–3, encoded)

- **Target variable**:
  - `target`: presence of heart disease (0 = no, 1 = yes)

In the implementation, the dataset is loaded as a CSV file using the `pandas` library. Basic inspection of the data shape, column types, and missing values is performed to verify that the dataset is correctly integrated and ready for preprocessing.

3 Data Preprocessing

Before applying any machine learning method, it is essential to ensure that the dataset is clean, consistent, and properly scaled. In this project, the preprocessing pipeline consists of three main steps:

1. handling missing values,
2. normalization of numeric features,
3. splitting the data into training and testing sets.

All preprocessing steps were implemented in Python using the `pandas` and `numpy` libraries. The corresponding code is organized into modular functions in the Jupyter notebook in order to keep the implementation clean and reusable.

3.1 Handling Missing Values

After loading the integrated heart disease dataset as a CSV file, we first inspected the data using `df.info()` and `df.isna().sum()` to check for missing values and potential inconsistencies. To ensure a complete dataset, we applied a simple but robust imputation strategy:

- For **numeric** features (such as `age`, `trestbps`, `chol`, `thalach`, and `oldpeak`), missing values were replaced with the *median* of the corresponding column.
- For **categorical or binary** features (such as `sex`, `cp`, `fbs`, `restecg`, `exang`, `slope`, `ca`, and `thal`), missing values were replaced with the *mode* (most frequent value) in each column.

Median imputation is less sensitive to outliers than mean imputation and is therefore suitable for medical measurements that may contain extreme values. Mode imputation for categorical variables preserves the most common category and avoids introducing unrealistic new values.

The imputation was implemented in a dedicated function `impute_missing_values()`, which takes the dataset and the lists of numeric and categorical columns as input and returns a cleaned version of the data. After this step, we verified that the number of missing values in all columns is zero.

3.2 Normalization of Numeric Features

Because the KNN classifier is a distance-based algorithm, it is crucial that all numeric features are on a comparable scale. Otherwise, attributes with larger numeric ranges would dominate the distance computation and bias the classification result.

In this project, we applied **Min–Max scaling** to all numeric features. For a numeric attribute x, the normalized value x' is computed as:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}}, \tag{1}$$

where $x_{\min}$ and $x_{\max}$ are the minimum and maximum values of that feature in the *training* set.

To avoid data leakage, the scaling parameters (minimum and maximum) were computed using only the training data. The same parameters were then applied to transform the test data. This was implemented in the function `min_max_scale_train_test()`, which returns scaled versions of both the training and test sets.

If a numeric column happened to be constant (i.e., $x_{\max} = x_{\min}$), the normalized values for that column were set to zero to avoid division by zero.

3.3 Train–Test Split

To evaluate the generalization performance of the KNN classifier, the dataset was split into a training set and a testing set. The data were first randomly shuffled and then divided using an **80% / 20%** split ratio:

- 80% of the instances were used for training the model,
- 20% of the instances were held out as an independent test set.

The split was implemented in a helper function `train_test_split_df()`, which takes the full dataset and a `test_size` parameter as input. A fixed random seed was used to ensure reproducibility of the results.

After the split and scaling, the final matrices `X_train`, `y_train`, `X_test`, and `y_test` were obtained. These were used in the subsequent experiments with the KNN classifier.

4 Exploratory Data Analysis

Before training the classifier, we performed an exploratory data analysis (EDA) in order to gain a better understanding of the feature distributions, potential outliers, and relationships between variables. All visualizations were generated in Python using the `matplotlib` and `seaborn` libraries.

4.1 Histograms of Numeric Features

To gain an initial understanding of the numerical attributes in the dataset, histograms were generated for the five key continuous features: `age`, `chol`, `oldpeak`, `thalach`, and `trestbps`. These plots provide insights into the overall shape of each distribution, the presence of skewness, and the frequency of extreme values. The following observations summarize the patterns apparent in the data.

Age. Figure 1 shows that the patients' ages range roughly from 30 to 80 years. The distribution is close to a quasi-normal shape with a clear peak around 55–60 years. Middle-aged and older adults constitute the majority of the dataset, while individuals younger than 40 appear far less frequently. This pattern is clinically expected, as the risk of heart disease increases markedly with age. A mild decline in frequency is observed beyond age 65–70, reflecting the lower proportion of very elderly patients in the dataset.

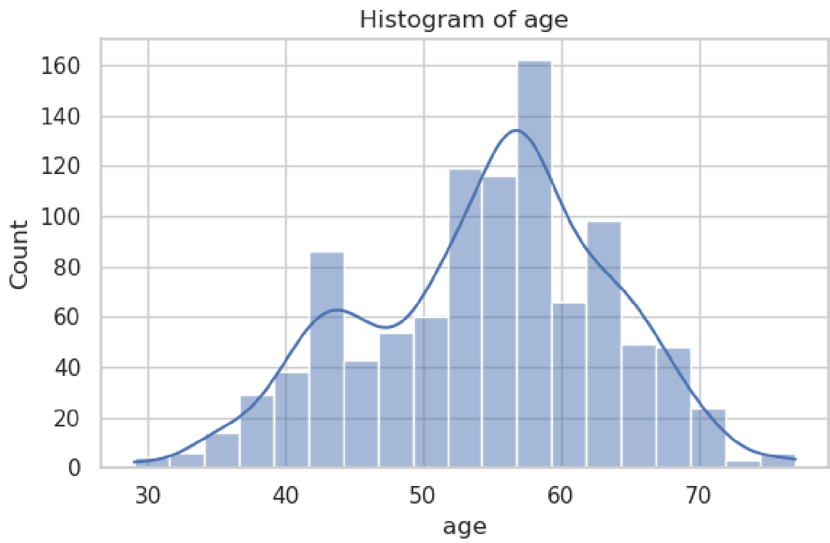

Figure 1: Histogram of the `age` attribute.

Cholesterol (`chol`). The histogram of `chol` (Figure 2) exhibits a moderately right-skewed distribution. The majority of values lie between 200 and 260 mg/dl, a range commonly associated with borderline or high cholesterol levels in clinical practice. A long upper tail extends to extreme values near 400–550 mg/dl, reflecting the presence of a few patients with severe lipid abnormalities. These points represent medically plausible outliers and are consistent with real-world cardiac risk factors.

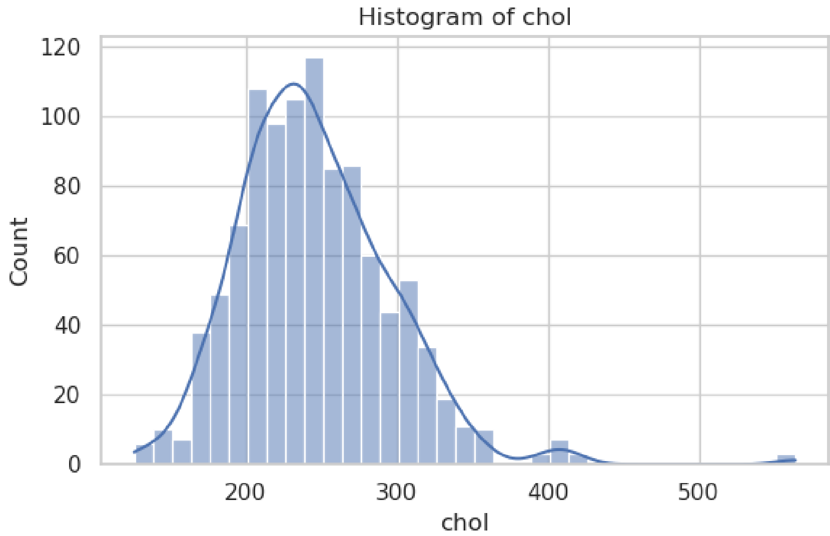

Figure 2: Histogram of the `chol` attribute.

Oldpeak. The feature `oldpeak`, representing ST depression induced by exercise, has the most heavily right-skewed distribution among all numeric attributes (Figure 3). A large proportion of patients have an `oldpeak` value close to zero, indicating minimal ST segment change. The frequencies decrease rapidly as the value increases, with only a small number of observations above 3, and rare extreme values reaching 5 or 6. This pattern aligns with clinical expectations, as significant ST depression is relatively uncommon in the general cardiac patient population.

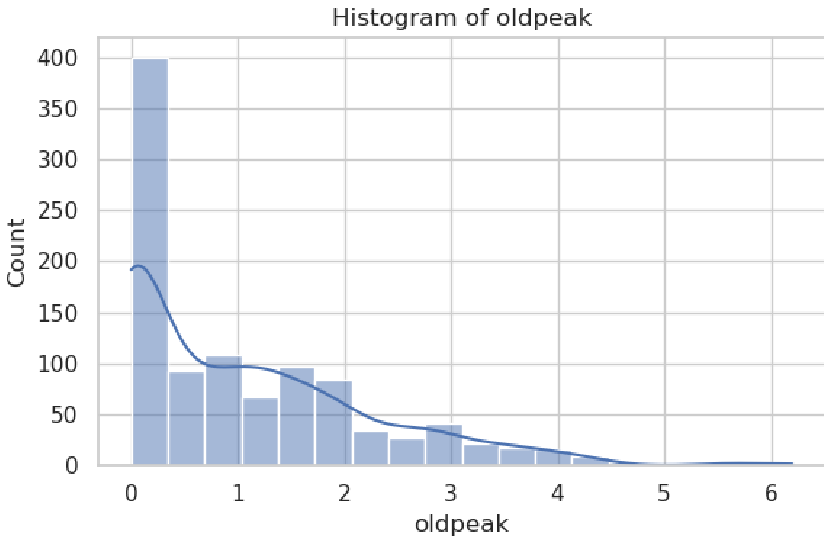

Figure 3: Histogram of the `oldpeak` attribute.

Thalach. The maximum heart rate (`thalach`) follows a nearly symmetric, bell-shaped distribution (Figure 4). Most patients achieve a heart rate between 150 and 170 beats per minute during stress testing. Extremely low values (below 100) are rare, suggesting possible severe cardiac conditions or incomplete exercise tests. This behavior of `thalach` is well aligned with clinical observations across stress test populations.

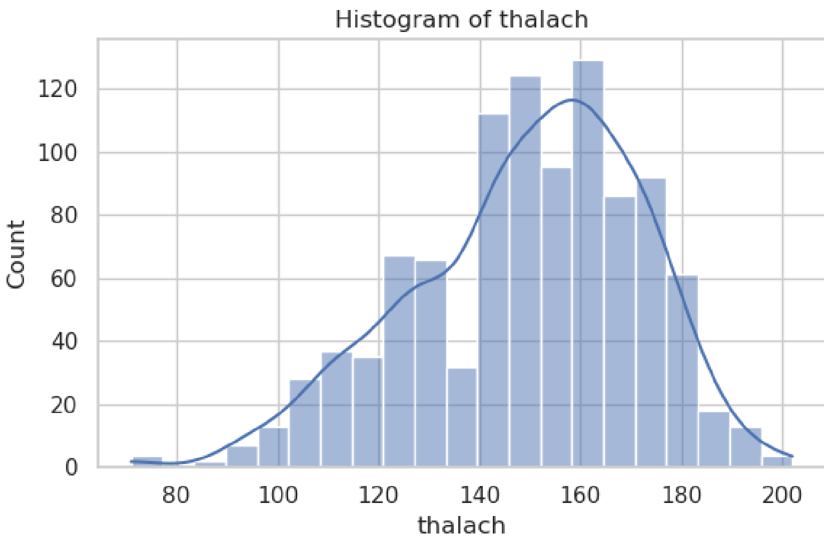

Figure 4: Histogram of the `thalach` attribute.

Resting Blood Pressure (`trestbps`). As shown in Figure 5, `trestbps` displays a distribution slightly skewed toward higher values. The central cluster of observations lies between 120 and 140 mmHg, corresponding to mild or moderate hypertension, a common risk factor for heart disease. Higher values up to approximately 180–200 mmHg appear infrequently but remain clinically realistic for untreated or high-risk patients.

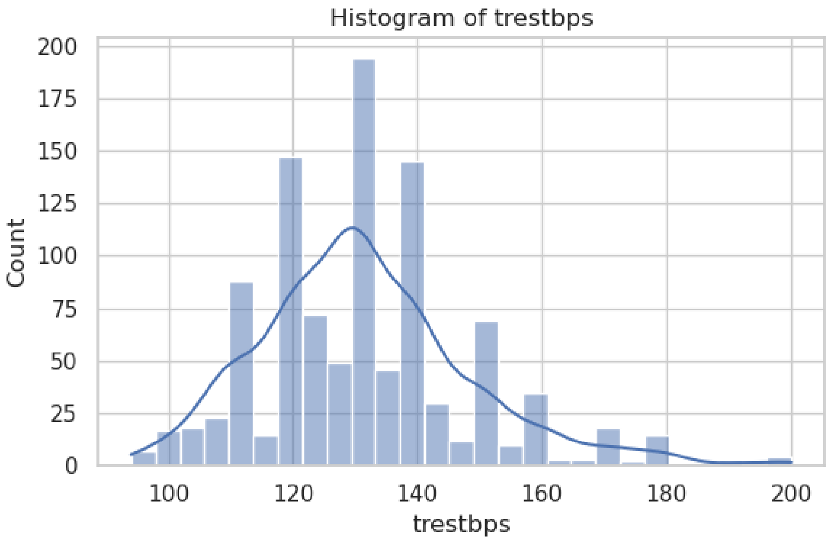

Figure 5: Histogram of the `trestbps` attribute.

Taken together, the histograms confirm that the numerical attributes exhibit a combination of near-normal, moderately skewed, and heavily skewed distributions, each consistent with real-world medical behavior. These observations further motivate the use of normalization before applying distance-based algorithms such as KNN.

4.2 Boxplots and Outliers

Boxplots were generated for the five main numeric features in order to examine their spread, central tendency, and potential outliers. These visualizations highlight the variability of each clinical measurement and provide insights into whether extreme values are present. The following describes the observations from each boxplot in detail.

Age. The boxplot of `age` (Figure 6) shows that the majority of patients fall between approximately 50 and 62 years of age, with a median around 56–58 years. No significant outliers are present, indicating a relatively consistent age distribution. This pattern aligns well with clinical expectations, as heart disease is most common among middle-aged and older adults.

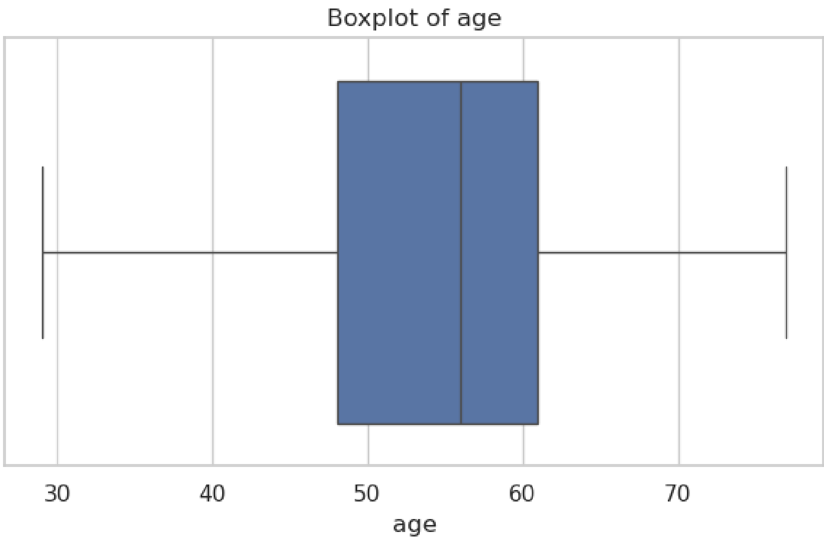

Figure 6: Boxplot of the `age` attribute.

Cholesterol (`chol`). The boxplot of `chol` (Figure 7) reveals a number of high-value outliers, with extreme cholesterol levels above 400 and even near 550 mg/dl. The interquartile range lies roughly between 200 and 275 mg/dl, and the median is around 240–250 mg/dl. These outliers are medically plausible, as severe hypercholesterolemia is a well-known risk factor for cardiac conditions.

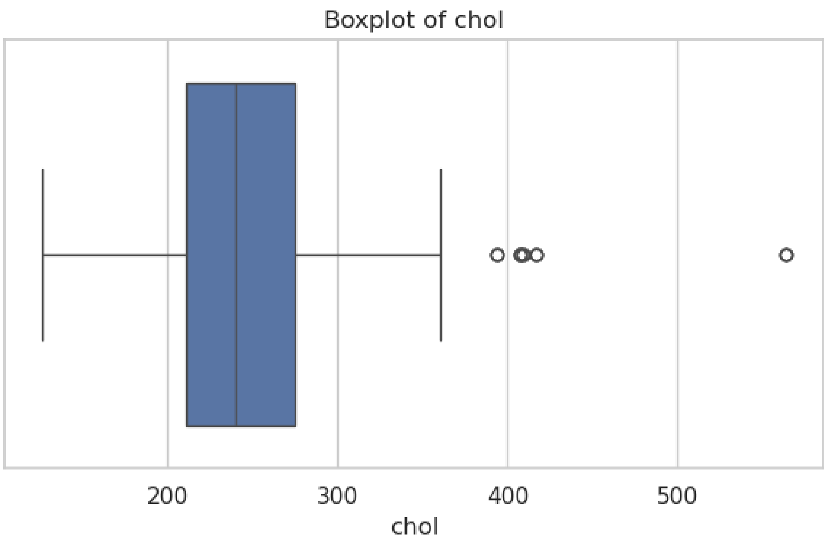

Figure 7: Boxplot of the `chol` attribute.

Oldpeak. The boxplot of `oldpeak` (Figure 8) shows a strongly right-skewed distribution, with the majority of values located between 0 and 1.5. A few substantial outliers appear above 5, indicating rare but clinically significant cases of severe ST depression during exercise testing. These extreme values contribute to the heavy-tailed structure observed in the histogram of this feature.

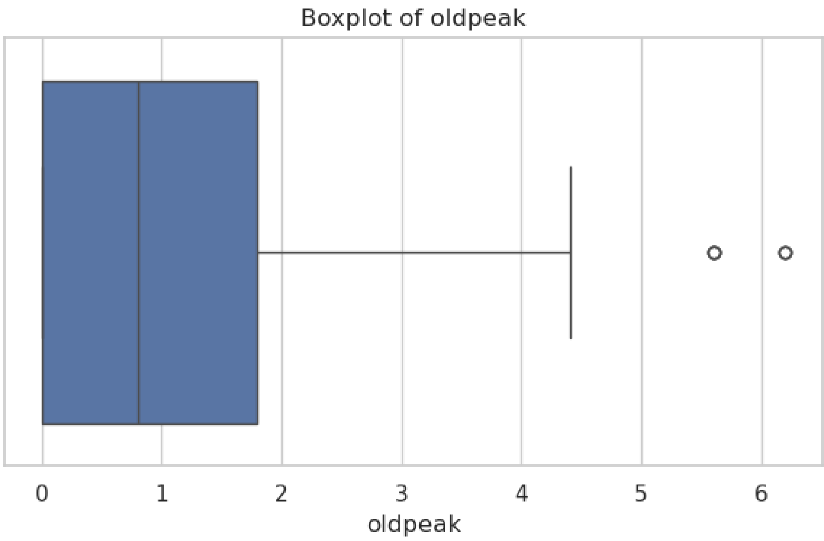

Figure 8: Boxplot of the `oldpeak` attribute.

Thalach. In the boxplot of `thalach` (Figure 9), the central mass of the data lies between approximately 140 and 170 beats per minute, with a median near 155 bpm. A single low outlier around 80 bpm is visible, which may correspond to severe cardiac limitations or an incomplete stress test. Upper values extend to about 200 bpm, which is physiologically plausible in stress testing scenarios.

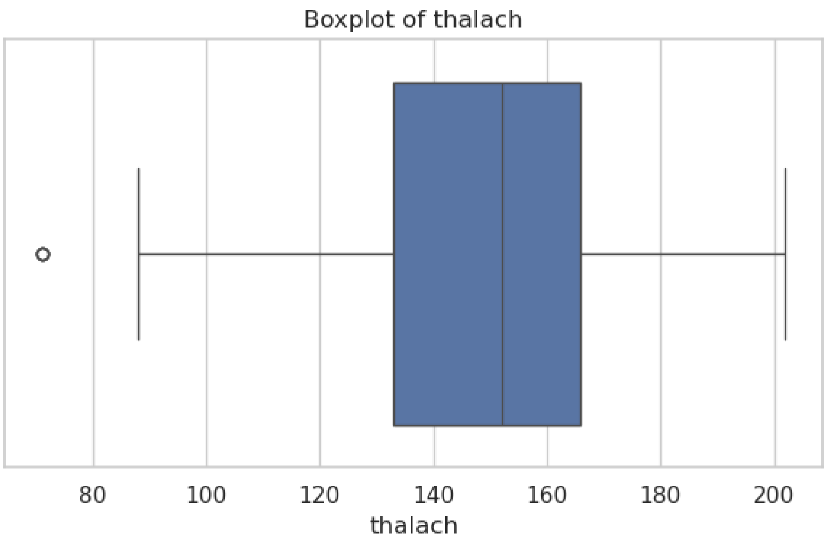

Figure 9: Boxplot of the `thalach` attribute.

Resting Blood Pressure (`trestbps`). The boxplot of `trestbps` (Figure 10) shows a typical clinical distribution, with most values in the 120–140 mmHg range and a median around 130–135 mmHg. Several pronounced outliers appear at 170–200 mmHg, indicating cases of severe hypertension. These observations are consistent with known risk patterns in cardiac patients.

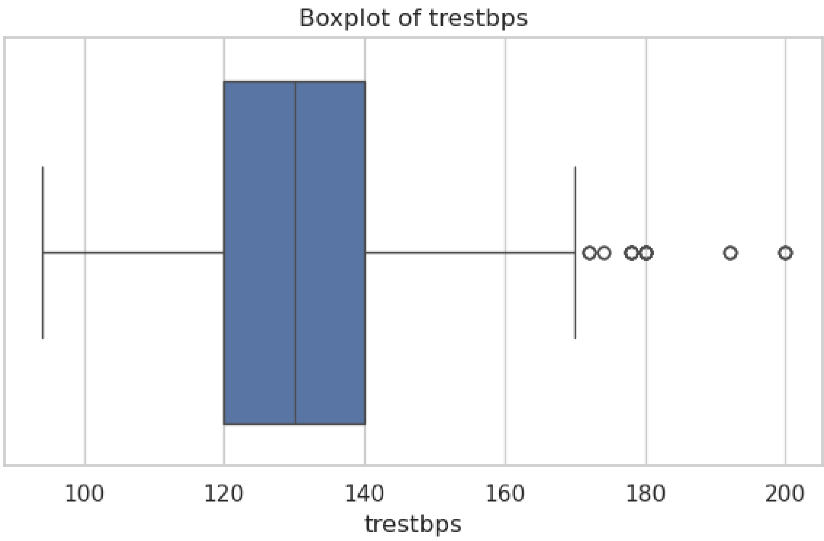

Figure 10: Boxplot of the `trestbps` attribute.

Overall, the boxplots confirm the presence of meaningful medical outliers in features such as `chol`, `oldpeak`, and `trestbps`, while attributes like `age` and `thalach` exhibit relatively stable distributions. Since KNN is generally robust to a small number of well-separated outliers and normalization was applied during preprocessing, these extreme values were retained rather than removed.

4.3 Q–Q Plots

To examine how closely the numeric features follow a normal distribution, Q–Q plots were generated for three representative attributes: `age`, `chol`, and `thalach`. In each plot, the sorted sample values are compared against the theoretical quantiles of a normal distribution with the same mean and standard deviation. Deviations from the diagonal line indicate departures from normality.

Age. The Q–Q plot for `age` (Figure 11) shows that most points lie close to the diagonal line, indicating that the central portion of the distribution is approximately normal. However, noticeable deviations appear in both tails: younger ages fall slightly below the reference line, while older ages (above roughly 75 years) lie somewhat above it. This pattern suggests a mildly heavy-tailed distribution but overall a fairly symmetric structure. Such behavior is typical for age in clinical datasets.

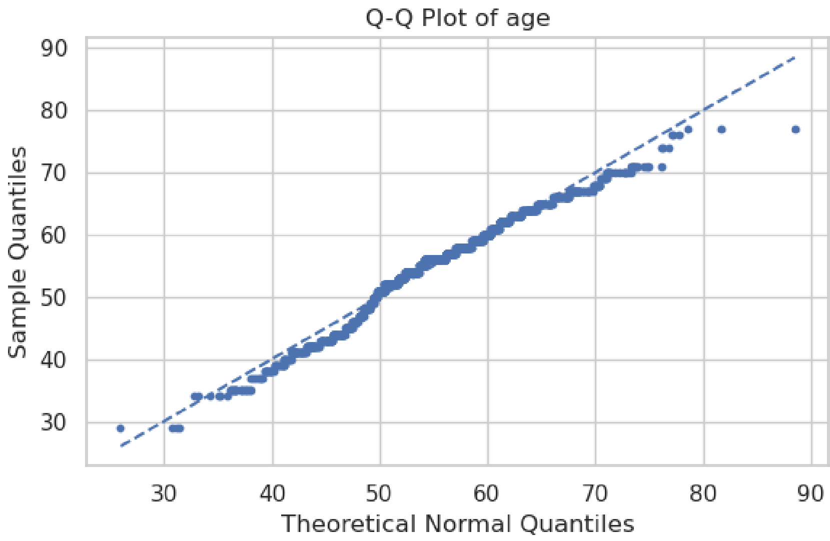

Figure 11: Q–Q plot of the `age` attribute.

Cholesterol (`chol`). The Q–Q plot of `chol` (Figure 12) reveals much stronger deviations, particularly in the upper tail. While the central region aligns reasonably well with the diagonal, several extreme values above 400 mg/dl rise sharply above the theoretical normal quantiles. These points correspond to clinically plausible cases of severe hypercholesterolemia. The resulting right-skewed behavior confirms that `chol` does not follow a normal distribution and contains meaningful outliers.

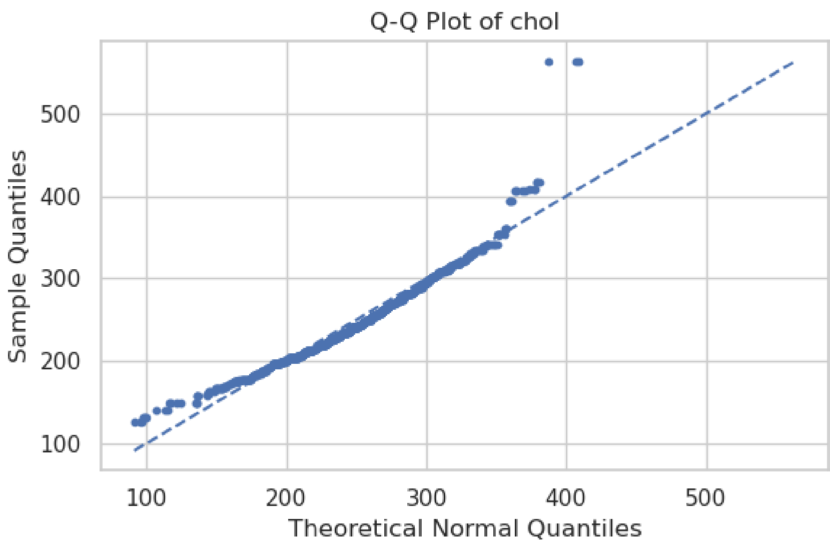

Figure 12: Q–Q plot of the `chol` attribute.

Thalach. The Q–Q plot of `thalach` (Figure 13) shows that the central portion of the distribution is approximately linear, indicating an almost normal pattern. However, the lower tail contains a few markedly low values (around 70–90 bpm), which fall well below the diagonal. Similarly, some values above 200 bpm appear slightly above the theoretical line. These deviations suggest mild skewness driven by rare low and high measurements, consistent with clinical variations observed in exercise testing.

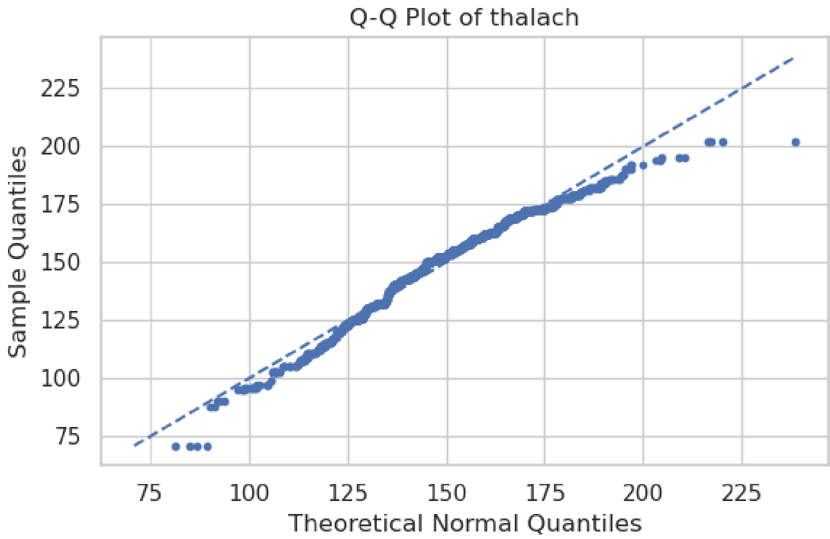

Figure 13: Q–Q plot of the `thalach` attribute.

Overall, the Q–Q plots confirm that none of the examined features follow a perfectly normal distribution. Each attribute exhibits tail deviations or skewness, with `chol` showing the strongest departure due to clinically meaningful high outliers. This is not problematic for the KNN classifier, which does not rely on normality assumptions.

4.4 Correlation Matrix

To investigate linear relationships among the numeric features and their association with the target variable, the Pearson correlation matrix was computed and visualized as a heatmap (Figure 14). The matrix reveals several clinically meaningful patterns and highlights which attributes are most relevant for distinguishing between patients with and without heart disease.

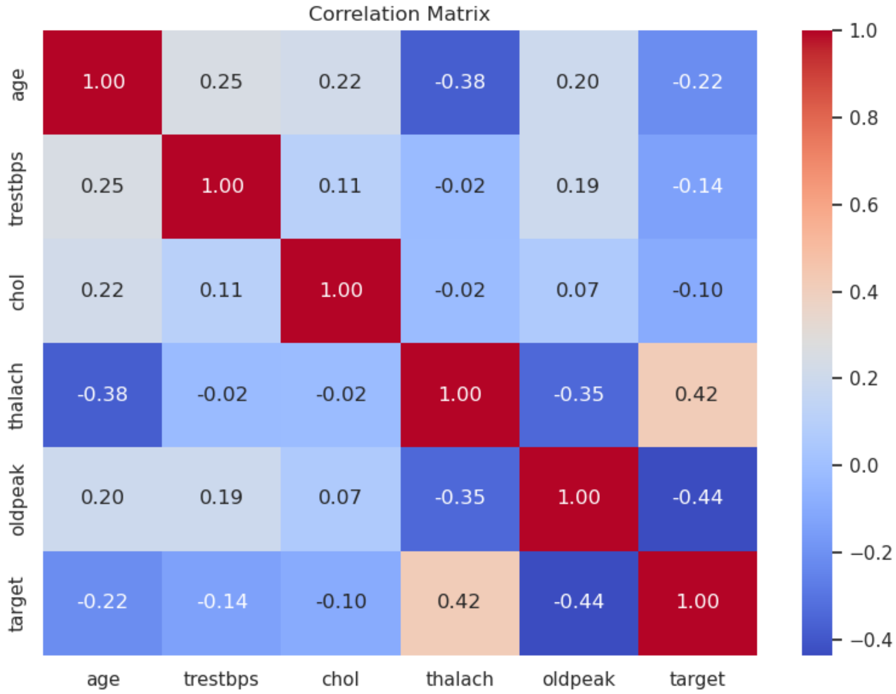

Figure 14: Correlation matrix of selected numeric features and the target variable.

The strongest correlations with the target variable are observed for `thalach` and `oldpeak`. The feature `thalach` (maximum heart rate achieved) shows a moderately positive correlation ($r = 0.42$), indicating that individuals who achieve higher heart rates during exercise are less likely to have heart disease. Conversely, `oldpeak` exhibits a moderately negative correlation with the target ($r = -0.44$), consistent with the clinical understanding that larger ST depression values are strongly associated with myocardial ischemia and a higher likelihood of disease. Age also shows a mild negative correlation with the target ($r = -0.22$), reflecting the increased prevalence of heart disease among older patients, although the relationship is not strictly linear.

Other features such as `trestbps` and `chol` display only weak correlations with the target, suggesting that—within this dataset—they contribute less predictive power on their own but may still play a role when combined with other variables in a multivariate context.

Several inter-feature relationships are also noteworthy. Age and `thalach` are moderately negatively correlated ($r = -0.38$), meaning that older individuals typically achieve lower maximum heart rates during stress testing. Likewise, `oldpeak` and `thalach` show a negative correlation ($r = -0.35$), reflecting the reduced exercise tolerance and performance typically seen in patients with ischemic responses.

Overall, the correlation matrix confirms that heart disease in this dataset is most strongly associated with exercise-related features such as `thalach` and `oldpeak`, while resting measurements such as `trestbps` and `chol` exhibit weaker linear dependencies with the target. These insights also highlight why distance-based models like KNN, which account for interactions among multiple features simultaneously, can still perform effectively despite the limited strength of individual linear correlations.

4.5 Pair Plots (Scatterplot Matrix)

To investigate multivariate relationships among the key numeric attributes, a pair plot was generated for `age`, `chol`, `thalach`, and `oldpeak`, with points colored according to the binary target variable. This visualization provides a simultaneous view of the marginal distributions of each feature as well as their pairwise interactions (Figure 15).

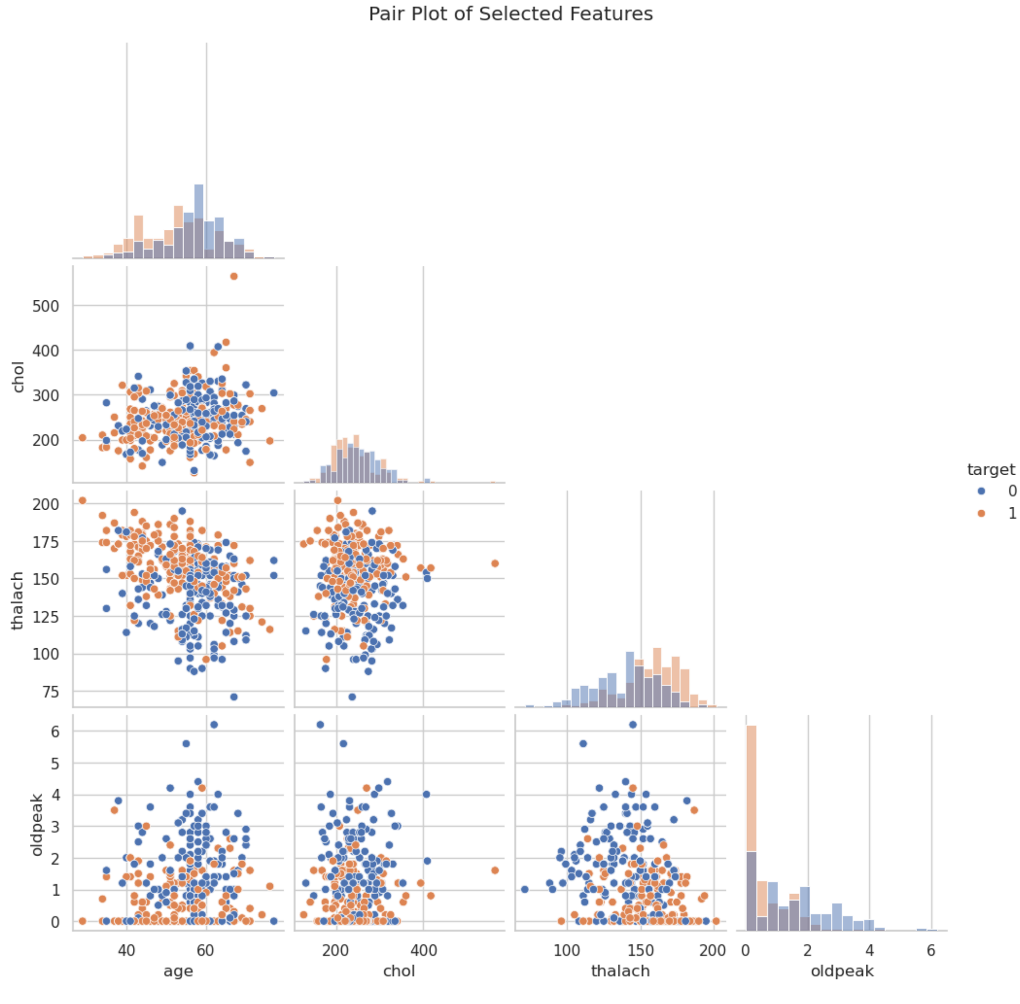

Figure 15: Pair plot for selected numeric features, colored by the target class.

The diagonal histograms show that `age` and `thalach` follow approximately symmetric distributions, whereas `chol` and `oldpeak` are noticeably right-skewed. In terms of class distribution, patients with heart disease (target = 1) appear slightly more often in higher age ranges, but the effect is modest. In contrast, clear class differences emerge in the distributions of `thalach` and `oldpeak`: individuals without heart disease typically achieve higher maximum heart rates, while those with heart disease tend to exhibit larger ST depression values.

The pairwise scatter plots reveal several important patterns. The relationship between `age` and `thalach` displays a clear negative trend reflecting the expected physiological decline in exercise capacity with age. Moreover, the combination of `thalach` and `oldpeak` provides the strongest visual separation between the two classes: non-diseased individuals cluster around high `thalach` and near-zero `oldpeak`, whereas diseased individuals tend to show lower `thalach` values and elevated `oldpeak`. In contrast, `chol` exhibits substantial overlap between the two classes in all pairwise projections, indicating that it is a weaker standalone discriminating feature.

Overall, the pair plots confirm that heart disease in this dataset is most strongly associated with exercise performance indicators (`thalach`, `oldpeak`), while features such as `chol` and `age` show weaker or less consistent separation. These multivariate patterns justify the use of a distance-based classifier such as KNN, which can exploit such relationships even when they are nonlinear or not clearly separable along any single dimension.

4.6 Count Plots of Categorical Features

To better understand the distribution of categorical variables in the dataset, count plots were generated for `cp` (chest pain type), `exang` (exercise-induced angina), `sex`, and the binary `target` variable. These plots illustrate the frequency of each category and help identify any potential imbalances in the dataset.

Chest Pain Type (`cp`). Figure 16 shows that chest pain type is far from uniformly distributed. The category `cp` = 0 (typical angina) is the most common with nearly 500 observations, while `cp` = 3 (asymptomatic) is the least frequent with fewer than 100 samples. The other two categories fall between these extremes. This distribution reflects real-world clinical observations, where typical angina is often the primary presenting symptom among cardiac patients.

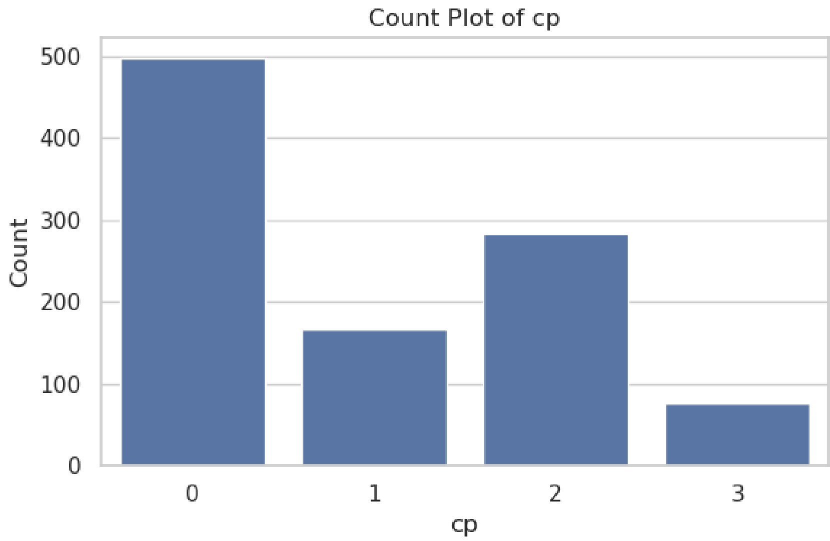

Figure 16: Count plot of the `cp` (chest pain type) attribute.

Exercise-Induced Angina (`exang`). As shown in Figure 17, the majority of patients (`exang` = 0) did not experience exercise-induced angina during stress testing. A smaller subset of roughly one-third of the dataset consists of patients who did experience angina (`exang` = 1). This imbalance is expected, as not all individuals display angina symptoms even when underlying disease is present.

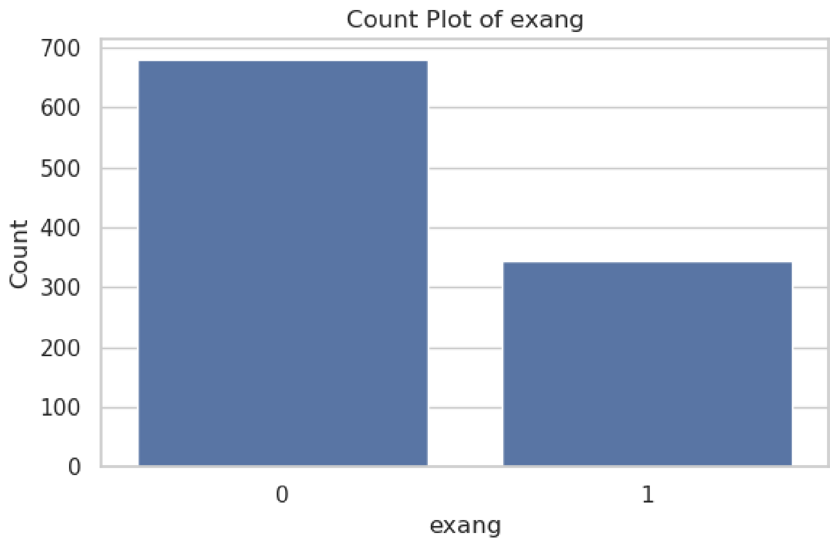

Figure 17: Count plot of the `exang` attribute.

Sex. Figure 18 demonstrates that the dataset contains a higher proportion of male patients (`sex` = 1) compared to female patients (`sex` = 0), with an approximate ratio of 2:1. This distribution reflects the higher prevalence of heart disease among men in many clinical populations.

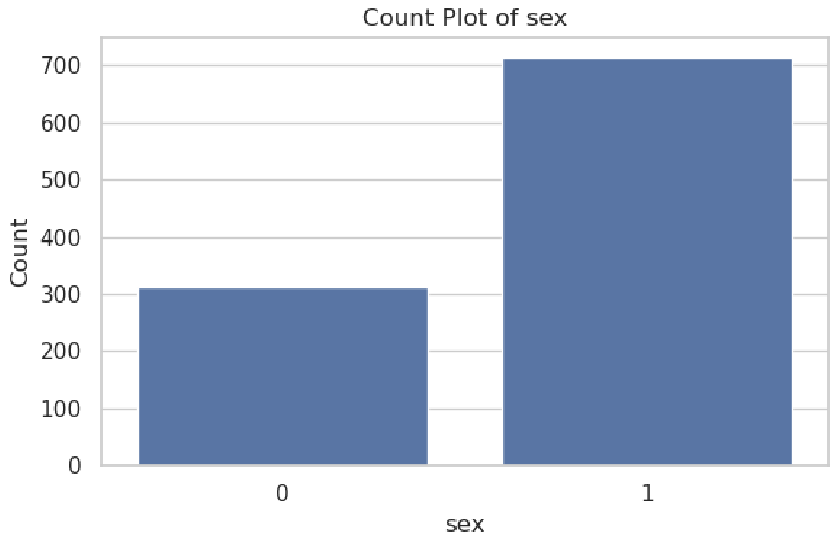

Figure 18: Count plot of the `sex` attribute.

Target Variable. The class distribution shown in Figure 19 indicates that the dataset is nearly balanced, with similar numbers of samples for `target` $= 0$ (no heart disease) and `target` $= 1$ (presence of heart disease). This balance is advantageous for training classification models, as it reduces the risk of bias toward either class and removes the need for rebalancing techniques such as oversampling or class weighting.

Figure 19: Count plot of the `target` variable.

Overall, the count plots reveal realistic clinical distributions across the categorical features. Although some categories—such as chest pain type and sex—exhibit notable imbalances, the balanced target distribution ensures that classification models such as KNN can be trained effectively without requiring adjustments for class imbalance.

5 KNN Classification

In this section, we apply the K-Nearest Neighbors (KNN) algorithm to classify patients based on their clinical measurements. KNN is an instance-based, non-parametric learning method and is particularly appropriate for this dataset because no strong linear separability was observed in the EDA, and several features (especially `thalach` and `oldpeak`) exhibit nonlinear relationships with the presence of heart disease. Since KNN relies directly on the distance between feature vectors, proper preprocessing and feature scaling are crucial components of the classification pipeline.

5.1 Overview of the KNN Algorithm

Given a test instance x, the KNN classifier operates by:

1. Computing the distance between x and all training samples,
2. Selecting the k closest training samples,
3. Determining the majority class among these k neighbors,
4. Assigning that class label to x.

Because KNN does not assume any particular probabilistic or functional form for the data, it is well suited for medical datasets where complex and nonlinear interactions exist among variables—as confirmed by the pair plots and correlation patterns observed in our EDA.

5.2 Mixed Distance Function for Numeric and Categorical Features

The dataset contains a mixture of numeric variables (such as `age`, `trestbps`, `chol`, `thalach`, and `oldpeak`) and categorical or binary attributes (e.g., `sex`, `cp`, `exang`, `restecg`). Using a single purely numeric distance metric is therefore inappropriate.

To address this, a mixed distance measure combining Euclidean distance for numeric features and Hamming distance for categorical features was implemented:

$$d(x, y) = w_{\text{num}} \cdot d_{\text{num}}(x, y) + w_{\text{cat}} \cdot d_{\text{cat}}(x, y),$$

where both weights were set to 0.5 in this study. This weighting ensures that numeric attributes— particularly influential ones such as `thalach` and `oldpeak`—contribute proportionally to the similarity computation, without overwhelming information provided by categorical features.

5.3 Model Training and Choice of k

After preprocessing and Min–Max normalization, the dataset was split into an 80/20 training–testing partition. The KNN classifier was trained on the resulting $\mathtt{X}_{\mathtt{train}}$ matrix and evaluated on $\mathtt{X}_{\mathtt{test}}$.

To examine the effect of the neighborhood size, we evaluated KNN for:

$$k \in \{3, 5, 7\}.$$

Performance was assessed using accuracy, precision, recall, and F1-score—metrics that are especially important in clinical contexts, where both false positives and false negatives carry significant consequences.

5.4 Classification Results

Table 1: Performance of the KNN classifier for different values of k.

k	Accuracy	Precision	Recall	F1-score
3	0.9317	0.9159	0.9515	0.9333
5	0.8195	0.7946	0.8641	0.8279
7	0.8341	0.8053	0.8835	0.8426

The highest accuracy was achieved at $k = 3$. This is consistent with the moderate level of class overlap observed in the scatter plots and pair plots: smaller values of k allow the classifier to better capture local patterns driven by the two most informative variables (`thalach` and `oldpeak`). Larger values of k tend to smooth out these local distinctions, causing a drop in performance.

Thus, the optimal neighborhood size for this dataset was:

$$k_{\text{best}} = 3.$$

5.5 Confusion Matrix

The confusion matrix for the best-performing model is shown in Figure 20.

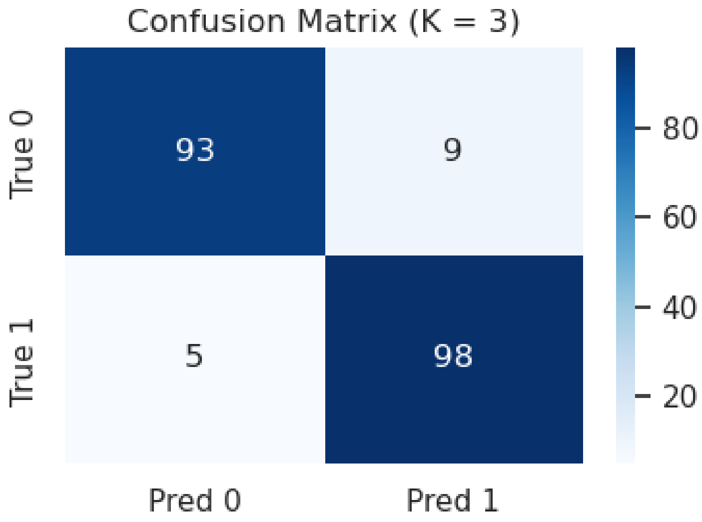

Figure 20: Confusion matrix of the KNN classifier for $k = 3$.

The classifier achieves a good balance between sensitivity and specificity. The relatively high recall indicates that the majority of heart disease cases were correctly identified. A small number of false positives and false negatives remain, which is expected given the overlapping distributions observed in the EDA—particularly for features such as `age` and `chol`, which show weak class separation. Nevertheless, the model overall demonstrates strong predictive performance given the mixed nature of the data and the limited number of highly discriminative features.

6 Conclusion

This project presented a data mining analysis of the integrated Heart Disease dataset from the UCI Machine Learning Repository, with the goal of predicting the presence of heart disease using a K-Nearest Neighbors (KNN) classifier. The workflow included data preprocessing, an extensive exploratory data analysis (EDA), and the implementation of a mixed-distance KNN model capable of handling both numeric and categorical attributes.

The preprocessing stage ensured that the dataset was suitable for distance-based learning. Missing values were imputed using median and mode strategies for numerical and categorical features, respectively. Numeric attributes were then normalized using Min–Max scaling to prevent features with large numerical ranges from dominating the distance computation. An 80/20 train–test split was used to evaluate the model on unseen data.

The EDA revealed several clinically meaningful patterns. Among all numeric attributes, `thalach` (maximum heart rate achieved) and `oldpeak` (exercise-induced ST depression) showed the strongest associations with heart disease. Patients with heart disease generally exhibited lower `thalach` values and higher `oldpeak` values, producing the clearest visual separation between classes in the scatter plots and pair plots. Other variables such as `chol` and `trestbps` exhibited weaker or more diffuse relationships and contained several medically plausible outliers. The target distribution was nearly balanced, which is favorable for classification without additional rebalancing.

The KNN classifier was implemented from scratch using a combined Euclidean–Hamming distance function. Model performance was evaluated for different values of k, and the best results were achieved at $k = 3$, where the classifier obtained the highest accuracy, precision, recall, and F1-score. This choice of neighborhood size aligns with the patterns observed in the EDA: because the data exhibit local structure driven by a few strongly discriminative features, a smaller value of k was best able to capture these local patterns without oversmoothing. The confusion matrix further showed that the classifier performed well in distinguishing between the two classes, correctly identifying most positive cases while maintaining a relatively low number of false alarms.

Overall, the study demonstrates that the KNN algorithm—despite its simplicity—can achieve strong predictive performance on this medical dataset when appropriate preprocessing and distance modeling are applied. The results highlight the importance of exercise-related features (`thalach` and `oldpeak`) as key indicators of heart disease, while also illustrating the limitations of features such as `chol` and `trestbps` when considered in isolation.

Future work could explore several extensions, including applying cross-validation for more robust estimation of model performance, optimizing or learning the numeric-to-categorical distance weighting, and comparing KNN with more advanced models such as logistic regression, decision trees, SVMs, or ensemble methods. Such approaches may further improve classification accuracy and yield additional insights into the relationships among features in cardiac risk prediction.

References

[1] UCI Machine Learning Repository: Heart Disease Data Set. `https://archive.ics.uci.edu/ml/datasets/heart+disease`